

AIVTS

Automotive Inventory Visibility and Tracking System

Comprehensive User Guide

Version 2.0 - Enterprise Edition

Covers: Installation . Data Import . Analysis . Reporting

Supported Formats: CSV . Excel . JSON . Parquet

Includes: Step-by-step walkthrough with test data

FAQ . Troubleshooting . Command reference

All data is stored locally in your browser. No server or internet required for core features.

This document is for informational purposes only.

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1. Overview

1.1 What is AIVTS?

The Automotive Inventory Visibility and Tracking System (AIVTS) is a fully standalone, zero-dependency web application designed for real-time automotive inventory management. It runs entirely in a web browser with all data stored locally - no server, no database, and no internet connection required for core operations.

AIVTS is distributed as a single HTML file (AutoInventoryTracker.html) that you open directly in any modern browser. There is nothing to install.

1.2 Core Modules

AIVTS includes thirteen fully integrated modules:

Description
Real-time stock visibility across all locations and KPIs
VIN-level vehicle movement and lifecycle management
SKU-level parts tracking and stock management
Shipment tracking with delay alerts
Physical vs. system stock verification
Low-stock alerts and purchase order management
Identification of slow-moving and dead stock
Damage tracking with insurance workflows
Purchase order and vendor performance analytics
Vehicle assignment to dealer bookings
Automated notifications and escalation workflows
BI dashboards with CSV export
Import company data for cross-organisational analysis
Access control and complete system audit trail

1.3 Key Features

- Single-file HTML app - no installation, no dependencies
- Four demo user accounts with different permission levels
- All data persisted in browser localStorage
- CSV export for vehicle data and audit logs
- Data Import module supporting CSV, Excel (XLSX/XLS), JSON, and Parquet formats
- Automatic column validation with helpful error messages

- Interactive data preview before committing an import
- Company analysis: industry distribution, revenue, employee metrics, geographic spread

2. Installation and Setup

2.1 System Requirements

AIVTS has minimal requirements:

Details
Chrome 90+, Firefox 88+, Edge 90+, Safari 14+
Windows, macOS, Linux, ChromeOS - any OS with a modern browser
~2 MB browser localStorage for data; ~150 KB for the HTML file
Not required for core features. Excel and Parquet parsing load a CDN library on first use.
None required

2.2 Opening the Application

Follow these steps to launch AIVTS:

- Step 1: Download or locate the file AutoInventoryTracker.html from the repository.
- Step 2: Double-click the file, or right-click and choose 'Open with' > your browser.
- Step 3: The AIVTS login screen will appear immediately - no loading screen or setup required.
- Step 4: If the browser shows a security warning about local files, click 'Allow' or 'Trust'.

? Note

If you see a blank page, ensure JavaScript is enabled in your browser settings.

Go to Settings > Privacy & Security > Site Settings > JavaScript > Allowed.

2.3 Demo Accounts and Login

AIVTS comes with four pre-configured demo user accounts. All accounts share the password: password

Access Level
Full access - View, Edit, Delete, Admin functions
Operational access - View and Edit
Stock and parts management - View and Edit
Read-only access to dashboards and reports

Select your account from the dropdown, enter the password 'password', and click Sign In. The main application dashboard will open.

3. Data Import Feature

The Data Import & Company Analysis module lets you load external company datasets into AIVTS for analysis. This is especially useful for analysing supplier portfolios, dealer networks, or market comparisons. The module supports CSV, Excel, JSON, and Parquet formats.

3.1 Accessing the Module

- Log in to AIVTS with any account.
- In the left sidebar, scroll to the 'Data' section.
- Click 'Data Import & Analysis'.
- The Data Import module will load in the main content area.

3.2 Supported File Formats

The module supports the following formats:

Offline?	
Yes	values. Fully offline. Recommended for most use cases.
CDN*	books. Reads the first sheet. Requires internet on first load.
Yes	ed object (see Section 3.3). Fully offline.
CDN*	mnar binary format. Requires internet on first load.

* CDN: The parsing library is loaded from the internet on first use only. Subsequent uses in the same session are offline.

3.3 Required Data Structure

Your data file must contain one row per company (or record), with each column representing a field. The first row must be the header row containing column names.

The following six columns are REQUIRED:

Description
Full name of the company
Industry or sector classification
Annual revenue (numeric, e.g. 3440000000 for \$3.44B)
Total number of employees (integer)
Year the company was founded (integer, e.g. 1945)
City, region, or country of headquarters

Additional optional columns (automatically detected if present): headquarters, website, ceo, stock_symbol, market_cap, description, phone, email, revenue_growth, profit_margin.

JSON Format Details

JSON files may be structured in any of the following ways:

```
// Option 1: Top-level array
[{"company_name": "Acme Corp", "industry": "Manufacturing", ...}]

// Option 2: Wrapped object with a 'companies' key
{"companies": [{...}, {...}], "metadata": {...}}

// Option 3: Wrapped object with a 'data' or 'records' key
{"data": [{...}, {...}]}
```

3.4 Column Naming Conventions

Column names are matched flexibly - spaces, underscores, and case are ignored. The table below shows all accepted variations for each required column:

Accepted Column Names
company_name, company name, name, company, organization, org_name
industry, sector, category, business_type, business type, vertical
revenue, annual_revenue, annual revenue, sales, turnover, income
employee_count, employees, headcount, staff, workforce, num_employees
founding_year, founded, year_founded, established, year established
location, city, headquarters, hq, region, country, place, base

Tip

Click the 'Column naming guide' link on the Data Import screen to view this table inside the application at any time.

3.5 Loading a File

To load a file:

- Navigate to Data Import & Analysis from the sidebar.
- Drag and drop your file onto the blue dashed upload zone, OR click the zone to open a file browser.
- Select a file with extension .csv, .xlsx, .xls, .json, or .parquet.
- AIVTS will automatically detect the format from the file extension.
- A loading indicator will appear while the file is being parsed.
- Once parsed, the Validation Results and Data Preview will appear below.

3.6 Validation Results

After loading, AIVTS checks your data against the required schema and displays a Validation Results panel. Each check is marked with one of three statuses:

Meaning
The column is found and contains valid data.
The column is found but may have data quality issues (e.g. non-numeric values in a numeric field). Import is still allowed.
A required column is missing or contains no data. You must fix this before importing.

The validation panel also shows the total row count and any optional columns detected. If all required columns pass, the 'Import & Analyse' button will be enabled.

3.7 Data Preview

Below the Validation Results, a Data Preview table shows the first 10 rows of your file. This lets you verify the data looks correct before committing to an analysis. Long cell values are truncated with an ellipsis (...) in the preview; hover over a cell to see the full value.

- Click 'Toggle Data Preview' to show or hide the preview table.
- Click 'Import & Analyse' to run the full analysis on all loaded rows.
- Click '? Load Another File' to start over with a different file.

3.8 Troubleshooting Load Errors

Common errors and solutions:

"Unsupported file type"

- Cause: You uploaded a file with an unsupported extension (e.g. .txt, .xml).
- Fix: Rename your file with a supported extension (.csv, .xlsx, .json, .parquet) or re-export it in a supported format.

"Invalid JSON"

- Cause: The JSON file contains syntax errors.
- Fix: Validate your JSON using an online tool like jsonlint.com. Ensure all keys are quoted and values

are properly formatted.

"Excel library not loaded"

- Cause: The SheetJS library could not be fetched from the CDN (no internet connection).
- Fix: Connect to the internet and try again, or convert your .xlsx file to .csv using Excel: File > Save As > CSV.

"Could not load Parquet parser"

- Cause: The hyparquet library could not be fetched from the CDN.
- Fix: Connect to the internet, or convert your Parquet file to CSV using Python: `pd.read_parquet('file.parquet').to_csv('file.csv', index=False)`

"Required column missing"

- Cause: A required column (e.g. company_name, revenue) was not found in the file.
- Fix: Check the Column Naming Guide (Section 3.4) and rename the column in your file to a recognised alias.

"All values are empty"

- Cause: A required column exists but contains only blank values.
- Fix: Populate the column in your source file before uploading.

"File is empty"

- Cause: The uploaded file contains no data or no header row.
- Fix: Open the file in a text editor or spreadsheet and verify it has at least one header row and one data row.

"Row column count mismatch (CSV)"

- Cause: Some CSV rows have a different number of commas than the header row.
- Fix: Open the CSV in a text editor. Look for rows with unescaped commas inside values. Wrap values containing commas in double quotes.

4. Running Analysis

4.1 Starting the Analysis

After successfully loading and validating a file, click the 'Import & Analyse' button in the Validation Results panel. AIVTS will process all rows and immediately display the full analysis dashboard below the preview section.

? Note

The 'Import & Analyse' button only appears when all required columns pass validation.
If you see only 'Toggle Data Preview' and '? Load Another File', check for ? Fail items.

4.2 Analysis Sections Explained

The analysis dashboard is divided into six sections, each giving a different perspective on the loaded dataset. All charts update dynamically based on the actual data in your file.

4.3 KPI Summary

The top row of the analysis shows four Key Performance Indicators (KPIs):

Description
Total number of records (rows) in the loaded dataset.
Count of distinct industry/sector values.
Mean annual revenue across all companies (formatted as \$M or \$B).
Mean employee count across all companies.

4.4 Industry and Geographic Distribution

Two horizontal bar charts are displayed side by side:

- Industry Distribution: Shows how many companies belong to each industry or sector, sorted from most to least common. The longest bar represents the dominant industry.
- Geographic Distribution: Shows how many companies are headquartered in each city or region (extracted from the location field). Only the top 8 locations are shown.

4.5 Revenue and Employee Distributions

Two more bar charts break down the dataset by size:

- Revenue Distribution: Bucketed into four ranges - < \$100M, \$100M-\$1B, \$1B-\$5B, and > \$5B. Each bar shows how many companies fall into that revenue bracket.
- Employee Scale: Bucketed into - < 1,000, 1K-10K, 10K-50K, and > 50K employees. Helps identify

the mix of SMEs vs. large enterprises in the dataset.

4.6 Top 5 Companies by Revenue

A ranked table lists the five highest-revenue companies in the dataset, showing their rank, name, industry tag, revenue (formatted), employee count, and founding year.

4.7 Key Insights

A text-based insights panel at the bottom of the analysis summarises the most important findings automatically, including:

- Total combined revenue across all companies
- Total combined workforce
- Dominant industry (most companies)
- Most common headquarters city
- Oldest and newest company by founding year
- Average founding year

5. Interpreting and Exporting Output

5.1 Reading the Charts

All charts in AIVTS use horizontal bar charts with consistent colour coding:

Meaning
Neutral data - industry counts, general distributions
Geographic distribution
Revenue distribution
Employee distribution
Alert or critical status items

Bar length is always proportional to the maximum value in that chart - the tallest/longest bar always fills 100% of the track width.

5.2 Exporting to CSV

You can export the full imported dataset (all columns, all rows) to a CSV file at any time after loading data:

- Run an analysis to ensure data is loaded.
- Click the '? Export CSV' button in the top-right of the Analysis card.
- A CSV file named 'AIVTS_ImportedData_YYYY-MM-DD.csv' will be downloaded automatically.
- This file contains all columns from the original uploaded file, with no modifications.

Export from other modules

The Reports & Analytics module offers a separate CSV export of vehicle inventory data (VIN, Model, Status, Location, Price). The User Roles module allows exporting the full system audit log to CSV.

5.3 Other Export Options

Currently available export formats in AIVTS:

Filename	
AIVTS_Vehicles_YYYY-MM-DD.csv	
AIVTS_ImportedData_YYYY-MM-DD.csv	(module)
AIVTS_AuditLog_YYYY-MM-DD.csv	

6. Other Modules: Inventory Management

6.1 Vehicle Tracking

The Vehicle Tracking module provides VIN-level tracking for all vehicles. Each vehicle record includes VIN, model, variant, colour, location, status, received date, and price. Status values include: Available, In Transit, PDI Pending, PDI Completed, Allocated, Sold, Damaged, and Delivered.

- Add vehicles via the '+ Add Vehicle' button.
- Filter by status, model, or location using the filter bar.
- Update vehicle status directly from the table.

6.2 Spare Parts Inventory

Tracks all spare parts by SKU with current stock quantity, minimum reorder level, unit cost, and supplier. Parts with stock below the minimum level are highlighted in amber.

- Add or edit parts records from the module.
- The Inventory Dashboard shows a 'Low Stock Parts' KPI in real time.
- Use Reorder Management to generate purchase orders for low-stock items.

6.3 Transit Tracking

Monitors all active shipments with origin, destination, carrier, expected and actual delivery dates, and current status (In Transit, Delayed, Delivered).

- Delayed shipments are highlighted in red.
- The Dashboard shows 'Active Shipments' and alerts for delayed items.

6.4 Reports and Analytics

The Reports module provides a comprehensive overview of the entire inventory system, including vehicle status breakdowns, parts value, operations summary, and supplier performance charts. Use the '? Export CSV' button to download the vehicle dataset.

7. Example Walkthrough Using Test Data

This section walks you through a complete end-to-end example using the provided test data files in the /test_data folder of the repository.

Step 1 - Open the Application

- Locate AutoInventoryTracker.html in the repository root.
- Double-click it to open in your browser, or drag it into a browser window.
- The AIVTS login screen appears.

Step 2 - Log In

- From the account dropdown, select 'Administrator (Full Access)'.
- Enter the password: password
- Click 'Sign In'.
- The Inventory Dashboard loads. (If this is your first visit, a quick tour will start.)

Step 3 - Navigate to Data Import

- In the left sidebar, scroll down to the 'Data' section.
- Click 'Data Import & Analysis'.
- The Data Import module loads with the upload zone visible.

Step 4 - Load the CSV Test File

- Click the blue dashed upload zone, or drag 'automotive_companies.csv' from the /test_data folder onto it.
- AIVTS will parse the CSV file (20 rows, 12 columns) instantly.
- A green toast notification confirms the file was loaded.

Step 5 - Review Validation Results

You should see the following validation summary (all green):

- Required column: 'company name' - 20 valid values ?
- Required column: 'industry' - 20 valid values ?
- Required column: 'revenue' - 20 valid values (numeric OK) ?
- Required column: 'employee_count' - 20 valid values (numeric OK) ?
- Required column: 'founding_year' - 20 valid values (numeric OK) ?
- Required column: 'location' - 20 valid values ?
- Row count - 20 data rows found ?
- Optional columns - Detected: headquarters, website, ceo, stock_symbol, market_cap, description ?

Step 6 - Preview the Data

The Data Preview table below the validation shows the first 10 rows. You can see all 12 columns including company_name, industry, revenue, employee_count, founding_year, location, headquarters, website, ceo, stock_symbol, market_cap, and description.

- Hover over any truncated cell to see its full value.
- Click 'Toggle Data Preview' to hide the preview if you want more screen space.

Step 7 - Import and Analyse

- Click the 'Import & Analyse' button (blue, below the validation results).
- The analysis dashboard will appear, scrolling into view automatically.
- A green 'Analysis complete ?' toast confirms success.

Step 8 - Interpret the Results

You should see the following results for the automotive_companies.csv dataset:

Expected Result
20
7 (Automobile Manufacturing, Two-Wheeler, Auto Components, etc.)
Approximately \$2.7B
Approximately 30,000
Auto Components (5 companies)
Chennai (5 companies)
Tata Motors / Mahindra & Mahindra (founded 1945)
Maruti Suzuki India (~\$4.1B)

Step 9 - Export the Data

- Click '? Export CSV' at the top-right of the analysis card.
- A file named AIVTS_ImportedData_YYYY-MM-DD.csv will download.
- Open it in Excel or a text editor to verify the full 20-row, 12-column dataset.

Step 10 - Try the Excel and JSON Files

Repeat steps 4-9 using automotive_companies.xlsx (requires internet for first load) and automotive_companies.json. All three files contain identical data and should produce identical analysis results.

Comparing formats

CSV: Fastest load, simplest format, fully offline.

Excel: Adds colour-coded header row and an Industry Summary sheet.

JSON: Includes metadata wrapper with source, version, and record_count fields.

8. Frequently Asked Questions

Q: Do I need to install anything?

A: No. AIVTS is a single HTML file. Open it in any modern browser. No installation, no server, no npm, no Python - nothing.

Q: Where is my data stored?

A: All inventory data (vehicles, parts, shipments, etc.) is stored in your browser's localStorage. Data never leaves your device. The Data Import module loads files directly in your browser memory and does not store them permanently.

Q: If I close the browser, will I lose my data?

A: No. Inventory records added via the AIVTS modules (vehicles, parts, etc.) are saved to localStorage and persist across sessions. Imported company data (from the Data Import module) is in-memory only and will need to be re-imported after a page refresh.

Q: Can I use AIVTS offline?

A: Yes, for all core inventory modules and for CSV/JSON imports. Excel and Parquet parsing requires a one-time internet connection to load the library. After loading once per session, the library stays in memory and no further internet is needed.

Q: Why does my Excel file not import?

A: Most likely the SheetJS library failed to load from CDN. Ensure you have an internet connection when first uploading an Excel file. Alternatively, open the file in Excel and save it as CSV, then upload the CSV version.

Q: My required columns are in the file but validation still fails. Why?

A: Check for invisible characters, extra spaces, or non-standard encoding in your column headers. The validator normalises spaces and underscores but cannot handle some Unicode whitespace variants. Copy-paste your header names from the Column Naming Guide table to be safe.

Q: How many rows can I import?

A: There is no hard limit, but performance may degrade above ~50,000 rows due to browser DOM rendering. For large datasets, consider pre-aggregating your data before import.

Q: Can I import data with different currencies?

A: The revenue column expects plain numeric values. If your file uses currency symbols (e.g. '\$3,440,000,000'), remove formatting before uploading: in Excel, select the column > Format Cells > Number (no symbol, no comma separator).

Q: How do I add more companies to my test dataset?

A: Edit automotive_companies.csv in a text editor or spreadsheet, add new rows below the header, and save. The file is plain text with comma-separated values.

Q: Can multiple users access AIVTS simultaneously?

A: AIVTS is a single-user browser application. Each user opens the HTML file independently on their own device. Data is not shared between devices unless you manually export and import it.

Q: How do I reset all data?

A: Click the '?? Clear All Data' button at the bottom of the left sidebar. This permanently deletes all records from localStorage. This cannot be undone.

Q: Is my data secure?

A: Data is stored only in your browser's localStorage and never transmitted anywhere. However, localStorage is accessible to any JavaScript running on the same origin. Do not store sensitive or regulated data in a production context without additional security controls.

Q: Can I export analysis charts to an image?

A: Not directly - AIVTS uses HTML/CSS bar charts rather than canvas-based charts. To capture a chart, use your browser's built-in screenshot tool (Windows: Win+Shift+S; Mac: Cmd+Shift+4; or browser DevTools > 'Capture screenshot').

Q: The page looks broken on my phone. Is mobile supported?

A: AIVTS is designed for desktop use (minimum ~1200px wide). On mobile, the layout may not display correctly. For best results, use a desktop or laptop browser.

9. Appendix

9.1 File Structure

Repository layout:

```
Automotive-Inventory-Visibility-and-Tracking-System/
?
??? AutoInventoryTracker.html    # Main application (open this)
??? README.md                   # Repository overview
?
??? test_data/
?   ??? automotive_companies.csv # 20-row CSV sample dataset
?   ??? automotive_companies.xlsx # Same data as Excel workbook
?   ??? automotive_companies.json # Same data as JSON
?   ??? README.md                # Test data field reference
?
??? docs/
    ??? AIVTS_User_Guide.pdf      # This document
```

9.2 Supported Column Name Reference (Quick Reference)

All Accepted Names

company_name, company name, name, company, organization, org_name, companyname
industry, sector, category, business_type, business type, vertical, segment
revenue, annual_revenue, annual revenue, sales, turnover, income, revenues, net_revenue
employee_count, employees, headcount, staff, workforce, num_employees, employee count, total_employees
founding_year, founded, year_founded, established, year established, incorporation_year
location, city, headquarters, hq, region, country, place, office, base

9.3 Converting Parquet to CSV (Python)

To convert a Parquet file for use with AIVTS offline:

```
import pandas as pd

# Read Parquet
df = pd.read_parquet('automotive_companies.parquet')
```

```
# Save as CSV
df.to_csv('automotive_companies.csv', index=False)

print(f'Converted {len(df)} rows to CSV')
```

9.4 Audit Log Actions Reference

The following action codes appear in the System Audit Log:

Description
User sign-in and sign-out events
Vehicle record created
Vehicle record modified
Vehicle record deleted
Spare part SKU created
Transit shipment record created
Alert marked as resolved
Vehicle data exported to CSV
Audit log exported
File loaded via Data Import module
Analysis run on imported data
Imported dataset exported to CSV

9.5 Keyboard Shortcuts and Tips

- Press Escape or click outside a modal dialog to close it without saving.
- Use Ctrl+F (Cmd+F on Mac) to search the browser page for specific values in tables.
- Use browser DevTools (F12) > Application > Local Storage to inspect or clear stored data directly.
- The sidebar can be scrolled vertically if your screen is shorter than the navigation list.
- All times displayed are in your local timezone (no UTC conversion needed).